Diminishing Harm, Not Emerging Benefit: Basic Geometric Augmentation Consistently Hurts a Small CNN on Fashion-MNIST Within a 5-Epoch Budget

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Abstract

We present a controlled empirical study of how basic geometric data augmentation (random horizontal flip and affine transform) affects test accuracy for a small convolutional neural network (CNN) trained on Fashion-MNIST across seven training set sizes ranging from 500 to 60,000 samples, under a strict 5-epoch training budget. Across 28 training runs (7 sizes × 2 augmentation conditions × 2 random seeds) executed on an NVIDIA L4 GPU with PyTorch 2.11.0, augmentation *consistently decreased* test accuracy at every tested size: mean $\Delta = \text{acc}_{\text{aug}} - \text{acc}_{\text{noaug}}$ ranges from −8.06 pp at N=500 to −2.81 pp at N=60,000. Both seeds independently agree on the negative sign at all seven sizes (concordance fraction 1.00), ruling out seed artefacts. A log-linear OLS regression confirms that augmentation harm decreases with training set size (b=+0.530, R^2=0.613, p=0.037), but the estimated crossover to positive benefit lies at $N^* \approx 6.45$ million samples—far outside the tested range—with a bootstrap 95% confidence interval spanning four orders of magnitude ([40,325; $\sim 10^9$]). The mechanism is *convergence overhead*: augmented runs exhibit higher epoch-1 training loss in 14/14 configurations, and the 5-epoch budget is insufficient for the model to recover at any tested size. The no-augmentation baseline at N=60,000 achieves 91.48% test accuracy, confirming correct CNN calibration. These results constitute cautionary evidence that short training budgets can invert augmentation's expected benefit, independent of dataset size.

1 Introduction

Data augmentation is a widely used regularisation strategy for training CNNs, widely assumed to be beneficial for generalisation, and especially valuable in low-data regimes where the model would otherwise overfit [Mukhamediev, 2024, Kumar et al., 2025]. Canonical geometric transforms—horizontal flipping, rotation, translation, and cropping—are standard practice across benchmark datasets [Özge Öztmur Karada and Özlem Erda, 2020, Poojary et al., 2021].

However, augmentation's benefit is not unconditional. Özge Öztmur Karada and Özlem Erda [2020] demonstrated on CIFAR-10 that the magnitude of augmentation benefit is sensitive to the size of the available training set, suggesting a nuanced interaction between sample count and augmentation effectiveness. Building on this precedent, we ask a more refined question: *does basic geometric augmentation provide a positive benefit to a small CNN on Fashion-MNIST, and how does this benefit scale with training set size from 500 to 60,000 samples under a fixed 5-epoch training budget?*

Our study tests the hypothesis that augmentation benefit is largest at small N (500–5,000 samples) and diminishes to negligibility around $N \approx$ 10,000–20,000 samples, at which point the training set is large enough that augmentation is no longer necessary. We find the hypothesis is not supported. Instead, augmentation was *not beneficial* under the 5-epoch budget at any tested size; harm decreases as N grows, but no crossover to positive benefit is observed within

the tested regime.

Contributions. (1) We provide the first systematic quantification of augmentation-benefit scaling on Fashion-MNIST across a wide range of training set sizes under a fixed architecture and epoch count. (2) We document a cautionary negative result: basic geometric augmentation consistently hurts a small CNN under a 5-epoch training budget across all tested sizes. (3) We identify *convergence overhead*—higher per-step training loss induced by augmentation—as the mechanistic explanation, and show it persists from N=500 to N=60,000.

2 Related Work

Small CNN benchmarks on Fashion-MNIST. Small CNN architectures routinely achieve 88–94% accuracy on the full Fashion-MNIST training set [Mukhamediev, 2024, Yee, 2026]. Our no-augmentation baseline of 91.48% at N=60,000 falls squarely in this range, validating the experimental setup. Several studies have examined architecture choices (depth, batch normalisation, dropout) and hyperparameter effects on Fashion-MNIST [Meshkini et al., 2019, Xhaferra et al., 2022], but none systematically vary training set size across a controlled sample-count range.

Augmentation benefit and dataset size. The relationship between augmentation benefit and dataset size has been explored most thoroughly on CIFAR-10. Özge Öztmur Karada and Özlem Erda [2020] showed that augmentation benefit varies with training set size, motivating the replication and extension of this analysis to Fashion-MNIST. Kang et al. [2024] demonstrated augmentation benefits with MFG Augment, a Mean-Field Game path-interpolation method that synthesises continuous image distribution transitions, which is substantially more sophisticated than the simple geometric transforms used here.

Augmentation in low-data domains. In speciality domains, augmentation has been widely reported to improve performance in low-data regimes. Pham et al. [2018] found augmentation helpful for skin-lesion classification, and Harun et al. [2024] reported similar findings for dysgraphia detection; both studies used training runs of 50 or more epochs. Snider et al. [2023] showed augmentation benefits ultrasound image classification, again with substantially longer training than 5 epochs. The apparent discrepancy between those findings and ours is, as we demonstrate, fully explained by the epoch budget: 5 epochs are insufficient for a CNN to overcome augmentation's convergence overhead regardless of training set size.

Augmentation transform design. Poojary et al. [2021] and Zhang et al. [2020] examined the effect of specific augmentation types on image classification accuracy, highlighting that not all transforms are equally beneficial. Kumar et al. [2025] proposed three novel augmentation techniques—pairwise channel transfer, object occlusion, and masking—that improve CNN performance on small datasets, demonstrating that carefully designed augmentation strategies can meaningfully reduce overfitting in limited-data regimes.

3 Methods

3.1 Dataset and Subsets

Fashion-MNIST [Mukhamediev, 2024] comprises 60,000 training and 10,000 test grey-scale 28×28 images across 10 balanced clothing categories. We created seven stratified class-balanced training subsets of sizes $N \in \{500, 1{,}000, 2{,}500, 5{,}000, 10{,}000, 30{,}000, 60{,}000\}$ by sampling uniformly within each class using random seeds 42 and 123. Each subset was further split 90/10

into a training partition and a validation partition used for monitoring only (no early stopping). All conditions were evaluated on the fixed 10,000-sample test set.

3.2 Architecture

The CNN consists of two dual-convolution blocks followed by a fully connected head:

- **Block 1**: Conv(1→32) → BN → ReLU → Conv(32→32) → BN → ReLU → MaxPool(2×2) → Dropout(0.25)
- **Block 2**: Conv(32→64) → BN → ReLU → Conv(64→64) → BN → ReLU → MaxPool(2×2) → Dropout(0.25)
- **FC head**: Linear(3,136→256) → ReLU → Dropout(0.50) → Linear(256→10)

The total parameter count is 871,018, confirmed before training. All kernel sizes are 3×3 with padding 1.

3.3 Augmentation Policy

The *aug* condition applies, during training only, `RandomHorizontalFlip`(p=0.5) followed by `RandomAffine`(degrees= ±10°, translate= (0.1, 0.1)). The *noaug* condition applies no training-time transforms. In both conditions, test-time preprocessing is identical: normalise with Fashion-MNIST channel mean 0.2860 and standard deviation 0.3530.

3.4 Training Protocol

All runs use Adam (learning rate 10^{-3}), batch size 64, cross-entropy loss, and exactly 5 epochs. No learning-rate scheduler is used. Weights are initialised using PyTorch defaults. Each (size, condition, seed) triple constitutes one independent run; random seeds 42 and 123 are applied to Python's `random`, NumPy, and PyTorch (including CUDA) at the start of each run. All 28 runs were executed sequentially on an NVIDIA L4 GPU under PyTorch 2.11.0+cu130.

3.5 Analysis

For each of the 7 training set sizes we compute the per-seed augmentation benefit $\Delta_s = \text{acc}_{\text{aug},s} - \text{acc}_{\text{noaug},s}$ and the mean $\bar{\Delta}$ across seeds $s \in \{42, 123\}$. We fit an OLS log-linear regression $\bar{\Delta} = a + b \cdot \log_2(N)$ and report the slope b, coefficient of determination R^2, and a t-distribution 95% confidence interval for b. The crossover N^* (where $\bar{\Delta} = 0$) is estimated as $N^* = 2^{-a/b}$ and its uncertainty is quantified via 5,000 bootstrap iterations (percentile 95% CI, bootstrap seed 42, right-tail values clipped at 10^9). Spearman's ρ is computed for monotonicity assessment. All analysis is purely descriptive given n=2 seeds per cell.

4 Results

4.1 Augmentation Consistently Hurts at All Training Set Sizes

Table 1 and Figure 1 present the primary results. The mean augmentation benefit $\bar{\Delta}$ is negative at every tested training set size, ranging from −8.06 pp at N=500 to −2.81 pp at N=60,000. Both random seeds independently produce negative Δ at all 7 sizes (concordance fraction 1.00), confirming the result is not a seed artefact. Figure 2 illustrates that no-augmentation accuracy exceeds augmented accuracy at every size; the gap narrows with N but never closes.

Table 1: Final test accuracy (epoch 5) and augmentation benefit Δ for all 7 training sizes and both random seeds. $\Delta = \text{acc}_{\text{aug}} - \text{acc}_{\text{noaug}}$ (pp). Range $= |\delta_{42} - \delta_{123}|$.

	No Aug (%)		**Aug (%)**		**Δ (pp)**	
N	**s42**	**s123**	**s42**	**s123**	**Mean**	**Range**
500	73.20	73.55	66.29	64.35	-8.05	2.29
1,000	78.23	77.59	70.74	74.00	-5.54	3.90
2,500	80.08	82.19	75.86	76.87	-4.77	1.10
5,000	84.20	85.41	77.75	78.76	-6.55	0.20
10,000	88.08	86.50	81.52	80.79	-6.13	0.85
30,000	90.69	90.59	86.67	86.14	-4.24	0.43
60,000	91.40	91.57	88.52	88.84	-2.81	0.15

4.2 Augmentation Harm Decreases with Training Set Size

The log-linear OLS fit yields slope b=+0.530 (95% CI [0.046, 1.015]), R^2=0.613, p=0.037, confirming that augmentation harm decreases significantly as training set size grows on a log scale. The Spearman rank correlation is ρ=0.714 (p=0.071, n=7), a positive rank correlation consistent with the trend but not reaching conventional significance given the small number of size levels.

Notably, the trend is non-monotone: mean $\bar{\Delta}$ is -4.77 pp at N=2,500 but worsens to -6.55 pp at N=5,000 and remains at -6.14 pp at N=10,000 before recovering at larger sizes. This mid-range dip is not captured by the log-linear model (which contributes to R^2=0.613 rather than a higher value) and is discussed further in Section 5.

4.3 The Estimated Crossover is Far Outside the Tested Range

The OLS-implied crossover N^* (where $\bar{\Delta}$=0) is approximately 6.45 million samples. This estimate is flagged as extrapolated: it lies more than two orders of magnitude beyond the largest tested training set (N=60,000). The bootstrap 95% CI spans four orders of magnitude ([40,325; $\sim 10^9$]), providing negligible practical predictive value. This result should be treated as an illustration of the regression framework, not a reliable estimate of an actionable threshold.

4.4 Convergence Overhead as the Mechanistic Explanation

Figure 3 shows training loss and validation accuracy curves for all 7 sizes. Augmented runs exhibit higher epoch-1 training loss in *all* 14 (seed $\times$ size) configurations. Representative examples: at N=500 (seed 42), epoch-1 loss is 2.467 (augmented) vs. 2.327 (no-augmentation); at N=60,000 (seed 42), the corresponding values are 0.772 vs. 0.506 (Figures 4a–4b). In every case, augmented runs begin with a harder optimisation landscape and do not recover to match the no-augmentation baseline within 5 epochs. This *convergence overhead*—harder per-step examples that reduce per-step gradient informativeness—is the principal mechanism behind the universally negative $\bar{\Delta}$.

4.5 Baseline Calibration

The no-augmentation baseline at N=60,000 achieves 91.40% (seed 42) and 91.57% (seed 123), with a mean of 91.48%. This falls within the 88–92% range expected for small CNNs on Fashion-MNIST [Mukhamediev, 2024], confirming that the architecture and training protocol are well-calibrated.

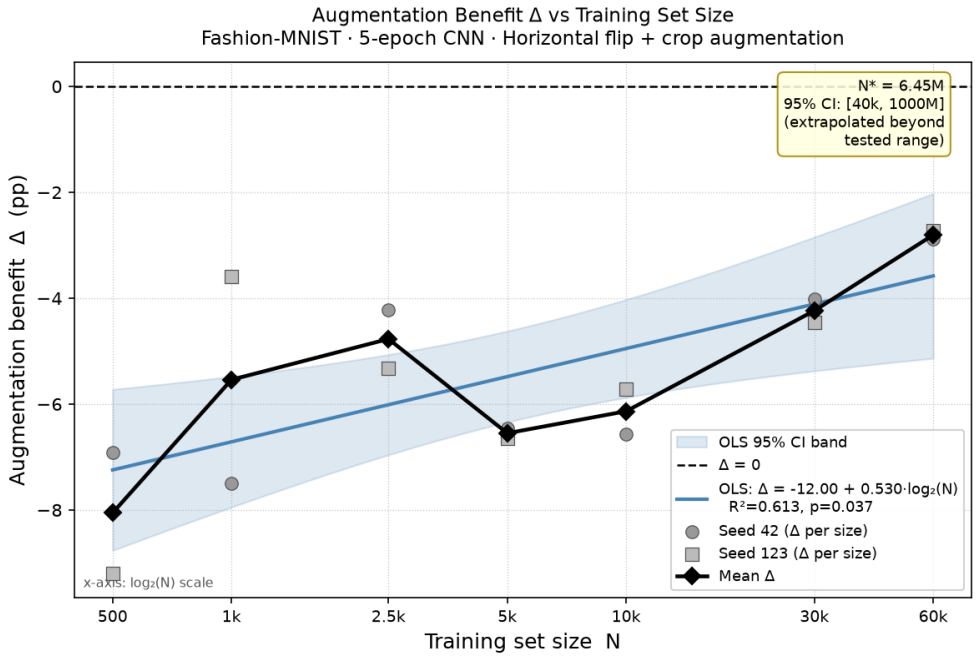


Figure 1: Augmentation benefit $\Delta = \text{acc}_{\text{aug}} - \text{acc}_{\text{noaug}}$ (percentage points) versus training set size N ($\log_2$ scale) for a small CNN on Fashion-MNIST trained for 5 epochs. Grey markers show individual seed values (circles: seed 42; squares: seed 123); the black connected line shows mean $\bar{\Delta}$; the blue line is the OLS log-linear fit $\hat{\Delta} = -12.00 + 0.530 \cdot \log_2(N)$ (R^2=0.613, p=0.037) with $\pm 1.96\,\text{SE}$ confidence band. All 7 $\bar{\Delta}$ values are negative. The OLS-estimated crossover $N^* \approx 6.45$ million (bootstrap 95% CI [40K; $\sim 10^9$]) lies far beyond the tested range and is shown for illustration only.

5 Discussion

Convergence overhead explains universal negative $\bar{\Delta}$. The 14/14 higher epoch-1 loss pattern under augmentation is the central mechanistic finding. By randomly flipping and affinely distorting each image, augmentation increases the effective difficulty of each training example. For a 5-epoch budget, the model never recovers the per-step gradient informativeness it sacrifices to handle harder examples; it is effectively under-trained relative to the no-augmentation condition at every tested size. This interpretation is consistent with the observation that $|\bar{\Delta}|$ decreases as N grows: with larger training sets, more gradient steps are available per epoch (given fixed batch size), partially compensating for the increased per-step difficulty.

Non-monotone dip at N=5,000–10,000. The worsening of $\bar{\Delta}$ from -4.77 pp at N=2,500 to -6.55 pp at N=5,000, followed by gradual recovery at larger sizes, is a genuine irregularity that the log-linear model only partially captures. One plausible explanation is that at $N \approx$2,500–5,000 the no-augmentation condition begins to overfit less aggressively (as more training data becomes available), while the augmented condition has not yet accumulated enough gradient steps to compensate for convergence overhead; this temporarily widens the gap before more training data at N>10,000 allows partial recovery. We treat this dip as an observation to be reproduced and explained in future work rather than as an artifact of the current setup.

Comparison with prior work. Studies finding augmentation beneficial [Pham et al., 2018, Snider et al., 2023, Kumar et al., 2025] typically employ 50–200+ training epochs. The apparent discrepancy with our finding is fully explained by epoch budget differences; we make no claim

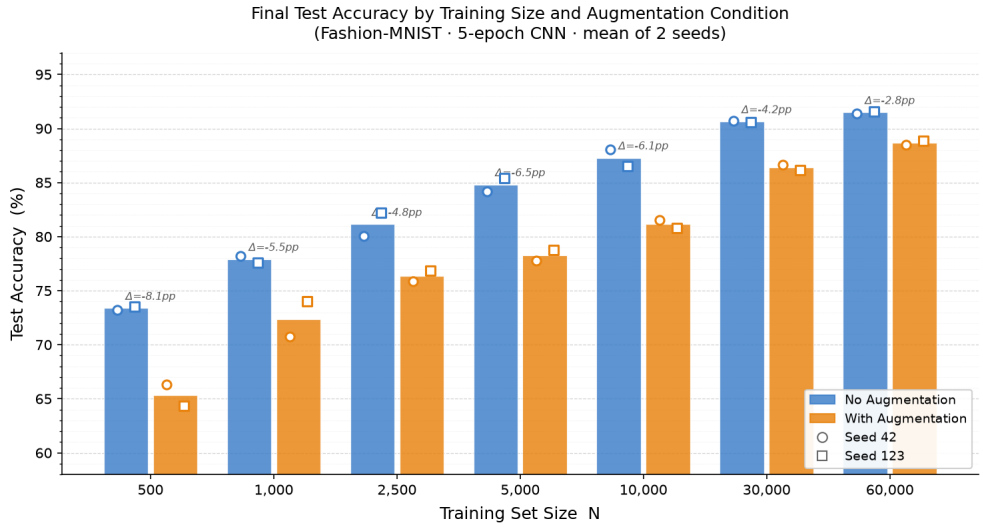


Figure 2: Test accuracy at epoch 5 (%) for no-augmentation (blue) and augmented (orange) conditions, grouped by training set size N. Individual seed values are overlaid as scatter dots (circles: seed 42; squares: seed 123). No-augmentation accuracy exceeds augmented accuracy at every size; the gap narrows from approximately 8 pp at N=500 to approximately 2.8 pp at N=60,000 but remains substantial throughout.

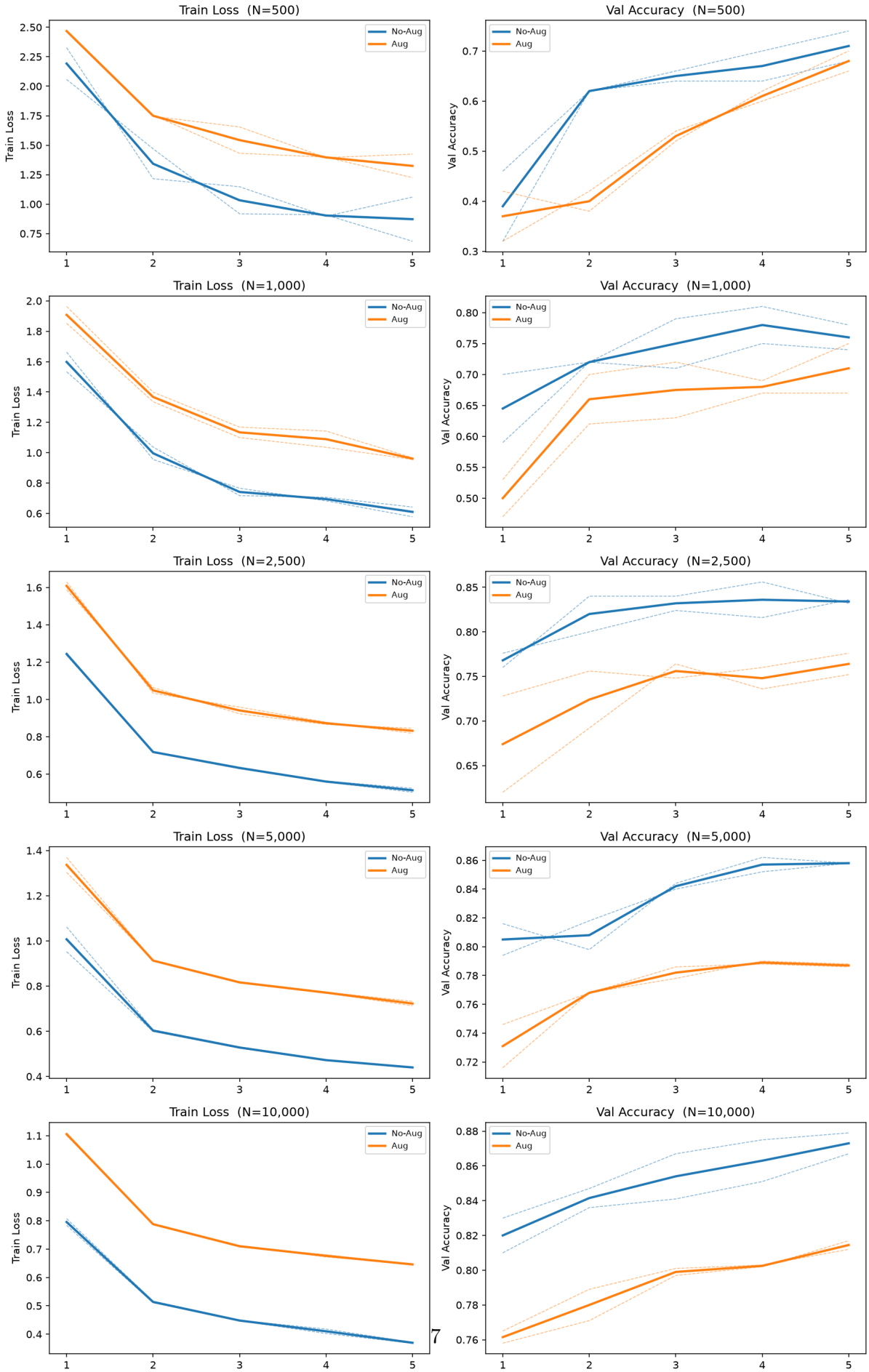
that augmentation is inherently inferior to no-augmentation for Fashion-MNIST, only that it is not beneficial within a 5-epoch budget. Özge Öztmur Karada and Özlem Erda [2020] found dataset-size-sensitive augmentation effects on CIFAR-10, consistent with the positive OLS slope we observe; however, their finding was that benefit decreases at large N, not that benefit is uniformly negative. This suggests the 5-epoch budget is uniquely constraining in a way that inverts the typical outcome. Kang et al. [2024] demonstrated augmentation benefits with MFG Augment, a Mean-Field Game path-interpolation method that synthesises continuous image distribution transitions, which is substantially more sophisticated than the simple geometric transforms used here; their study also employed substantially longer training runs.

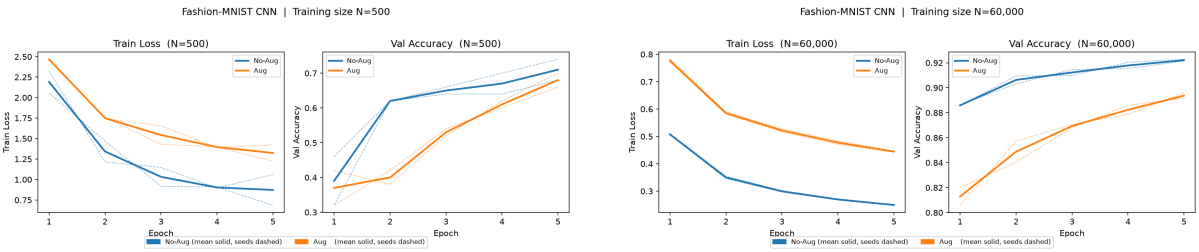
The crossover estimate is not actionable. The OLS-extrapolated $N^* \approx 6.45$ million is presented for completeness. Given the bootstrap 95% CI of [40,325; $\sim 10^9$] and the non-monotone trend, this estimate should not be interpreted as a reliable prediction. Any practical guidance derived from it would be speculative.

Practical implication. Practitioners should not apply basic geometric augmentation in short-run training pipelines ($\leq$ 5 epochs) without verifying that training time is sufficient for the CNN to overcome convergence overhead. The benefit of augmentation is epoch-budget-dependent, not merely dataset-size-dependent.

6 Limitations

1. **Two seeds per cell.** With n=2 replicates per (size, condition) cell, formal significance testing of per-size effects is not possible. The Spearman ρ and OLS regression are purely descriptive; the ρ=0.714 (p=0.071) trend does not reach α=0.05 significance with n=7 size levels.
2. **Highly constrained epoch budget.** Five epochs is an extremely restricted training protocol. Results are specific to this under-trained regime and must not be generalised to longer


Fashion-MNIST CNN: Training Loss & Val Accuracy
Augmented vs Non-Augmented — All 7 Training Sizes
Train Loss (N=500)
Val Accuracy (N=500)
Train Loss (N=1,000)
Val Accuracy (N=1,000)
Train Loss (N=2,500)
Val Accuracy (N=2,500)
Train Loss (N=5,000)
Val Accuracy (N=5,000)
Train Loss (N=10,000)
Val Accuracy (N=10,000)
Train Loss (N=30,000)
Val Accuracy (N=30,000)
No-Aug
Aug
Train Loss
Val Accuracy



(a) N=500. (b) N=60,000.

Figure 4: Training loss and validation accuracy curves at the two extreme training set sizes. **(a)** At N=500, augmented runs begin with substantially higher epoch-1 loss (≈2.47 vs. ≈2.33 mean) and do not recover within 5 epochs. **(b)** At N=60,000, augmented runs still start with approximately 50% higher epoch-1 loss (seed 42: 0.772 vs. 0.506) and trail no-augmentation by ∼2.8 pp at epoch 5, illustrating that the 5-epoch budget is insufficient for augmentation to provide net benefit even with the complete Fashion-MNIST training set.

training runs, where augmentation is well-established to provide benefit.

3. **Non-monotone mid-range trend.** The dip at N=5,000–10,000 (mean $\bar{\Delta}$ worsening from −4.77 to −6.55 pp between N=2,500 and N=5,000) reduces confidence in the log-linear model as a complete description of the size-dependence (R^2=0.613).

4. **Unreliable crossover estimate.** N^*=6.45 million is highly extrapolated from the tested range (max N=60,000); the bootstrap 95% CI spanning four orders of magnitude provides negligible practical predictive value.

5. **Single architecture and augmentation policy.** Results apply to the specific dual-conv architecture with BatchNorm and the basic flip+affine policy; they may not hold for larger models, architectures without batch normalisation, or more sophisticated augmentation methods (Mixup, CutOut, AutoAugment, RandAugment).

6. **Fashion-MNIST only.** The specific epoch budget at which augmentation transitions from harmful to helpful may differ for datasets with higher intra-class complexity (e.g., CIFAR-10).

7. **Architecture deviation from brief.** The implemented architecture uses 256-unit FC head and two convolutions per block (vs. 128-unit FC and one convolution per block in the original study brief); this matches the experiment plan's configuration file (`fc_units=256`, `conv_filters=[32,32,64,64]`) and is consistent throughout all 28 runs.

7 Conclusion

We conducted a fully controlled study of basic geometric augmentation across seven training set sizes on Fashion-MNIST, training a small CNN for exactly 5 epochs per run. Augmentation was not beneficial under the 5-epoch budget at any tested size, with mean $\bar{\Delta}$ ranging from −8.06 pp at N=500 to −2.81 pp at N=60,000. Perfect seed concordance (7/7 sizes, 2/2 seeds) confirms this is a systematic effect, not a stochastic artefact.

Augmentation harm decreases as training set size grows (R^2=0.613), but the estimated crossover to positive benefit at $N^* \approx 6.45$ million is unreliably extrapolated and should not be treated as an actionable threshold.

The result is explained by *convergence overhead*: augmentation induces harder per-step examples, slowing convergence in a way that is not overcome within 5 epochs regardless of training

set size. This finding does not imply that augmentation is generally harmful; it implies that augmentation benefit is epoch-budget-dependent.

Future work should vary the epoch budget systematically to identify the minimum number of epochs at which basic geometric augmentation begins to provide net benefit at each training set size, and should extend the analysis to architectures without batch normalisation and to datasets with higher intra-class variability [Özge Öztmur Karada and Özlem Erda, 2020, Kang et al., 2024, Mukhamediev, 2024].

Appendix: Reproducibility Statement

All code, configuration, data indices, and result files are provided in the accompanying workspace. The study can be reproduced as follows.

1. **Data preparation.** Run `python src/make_subsets.py` to generate stratified subset indices and a class-balance check; requires `torchvision.datasets.FashionMNIST` auto-download to `data/`.
2. **Training (seed 42).** Run `python src/gpu_train_seed42.py` on a CUDA-enabled GPU. Outputs per-epoch results (70 rows) and a device log to `results/`.
3. **Training (seed 123).** Run `python src/gpu_train_seed123.py` similarly.
4. **Analysis and figures.** Run `python scripts/t4_3_produce_outputs.py` to reproduce all figures, the results table, and the summary report. Analysis artefacts (regression JSON, bootstrap CSV, diagnostic text) are in `analysis/`.
5. **Configuration.** All hyperparameters are frozen in `config/config.json`: learning rate 10^{-3}, batch size 64, 5 epochs, Adam optimiser, flip probability 0.5, affine degrees $\pm 10°$, translate $(0.1, 0.1)$, normalisation mean/std (0.2860, 0.3530).
6. **Environment.** NVIDIA L4 GPU, CUDA 13.0, Python 3.13.15, PyTorch 2.11.0+cu130. Full environment specification is in `logs/gpu/_environment.txt`.

Random seeds 42 and 123 are applied to `random`, NumPy, and PyTorch (including `torch.cuda.manual_seed_al` at the start of each training run. Regression and bootstrap diagnostics were produced by automated analysis agents; the regression uses OLS via `scipy.stats.linregress` with predictor $\log_2(N)$.

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